Find Bugs Faster through ViewModel and API Library Testing

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https://www.orlandocodecamp.com/

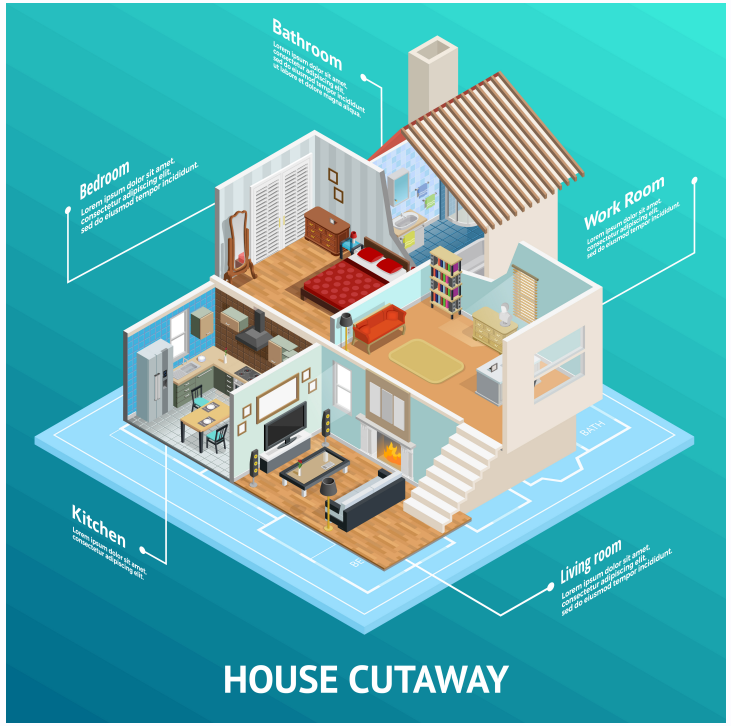
Topics

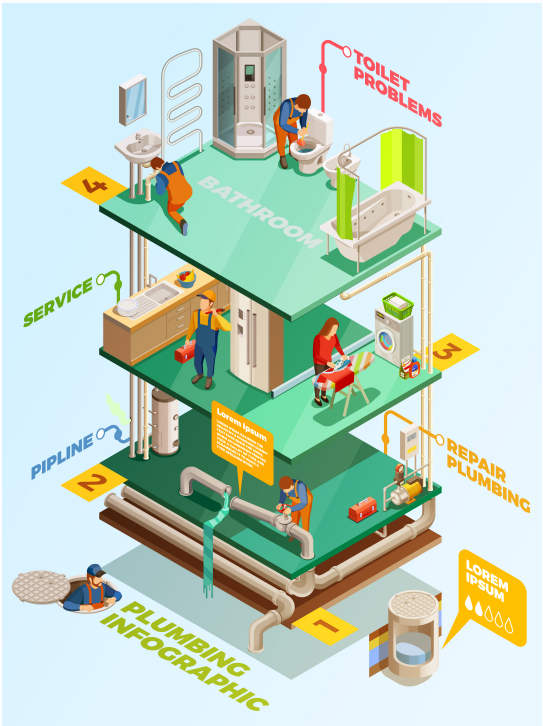
- What makes cross-platform and mobile different
- Real-world mobile design components
- Separating API consumption into testable libraries
- Testing how data flows through your ViewModels

Part 1

Imagine your house was an app ...

Perspectives on Data







User | Web Site | Mobile App

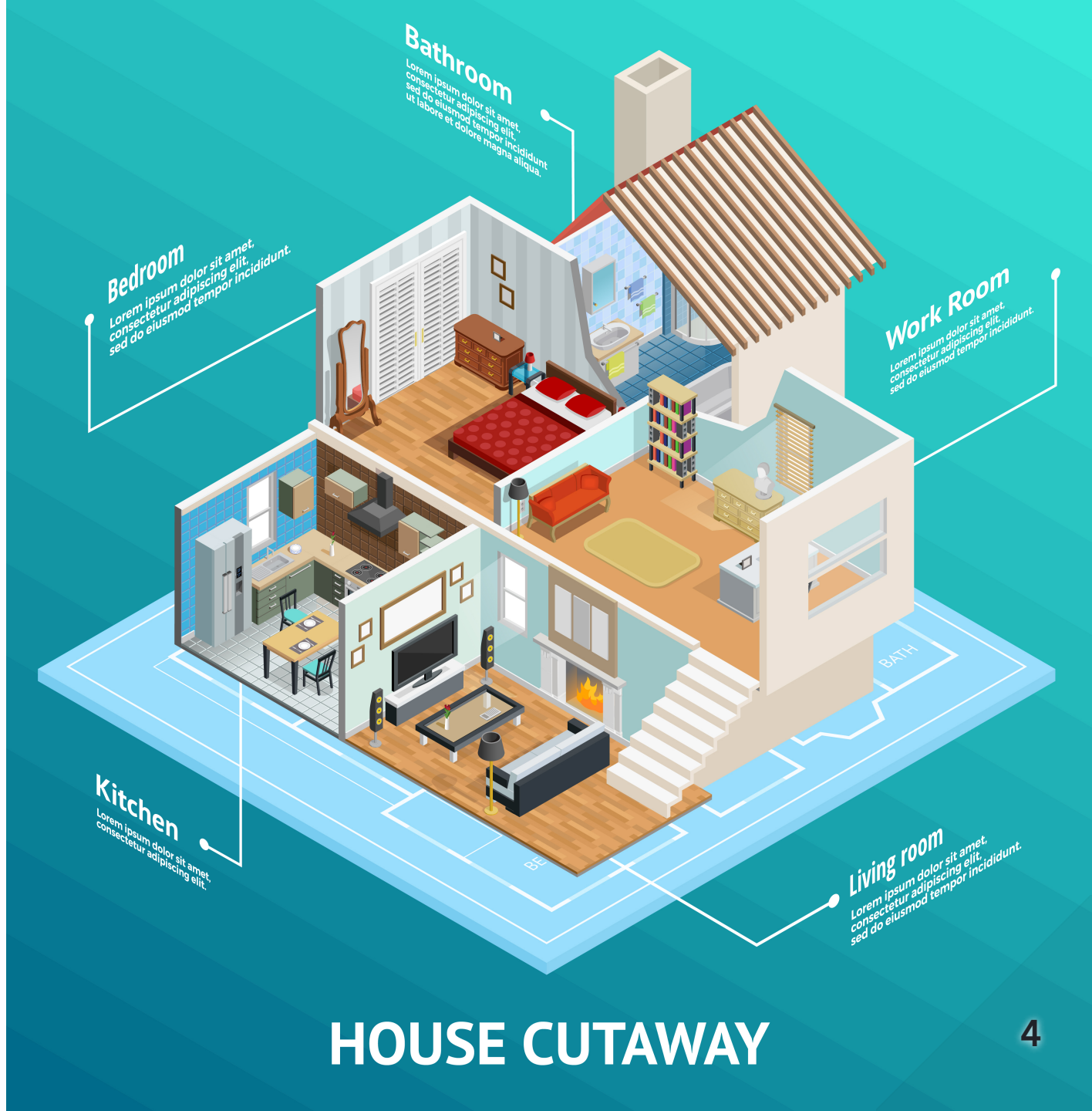
Images by macrovector on Magnific

Energy - User

[Web Site or Mobile App]

- Energy Consumption
  - Just Works
- Energy Production
  - Somebody Else's Problem
- Resiliency
  - Generator (Maybe?)
- **Focus**
  - **Living My Best Life**

Image by macrovector on Magnific

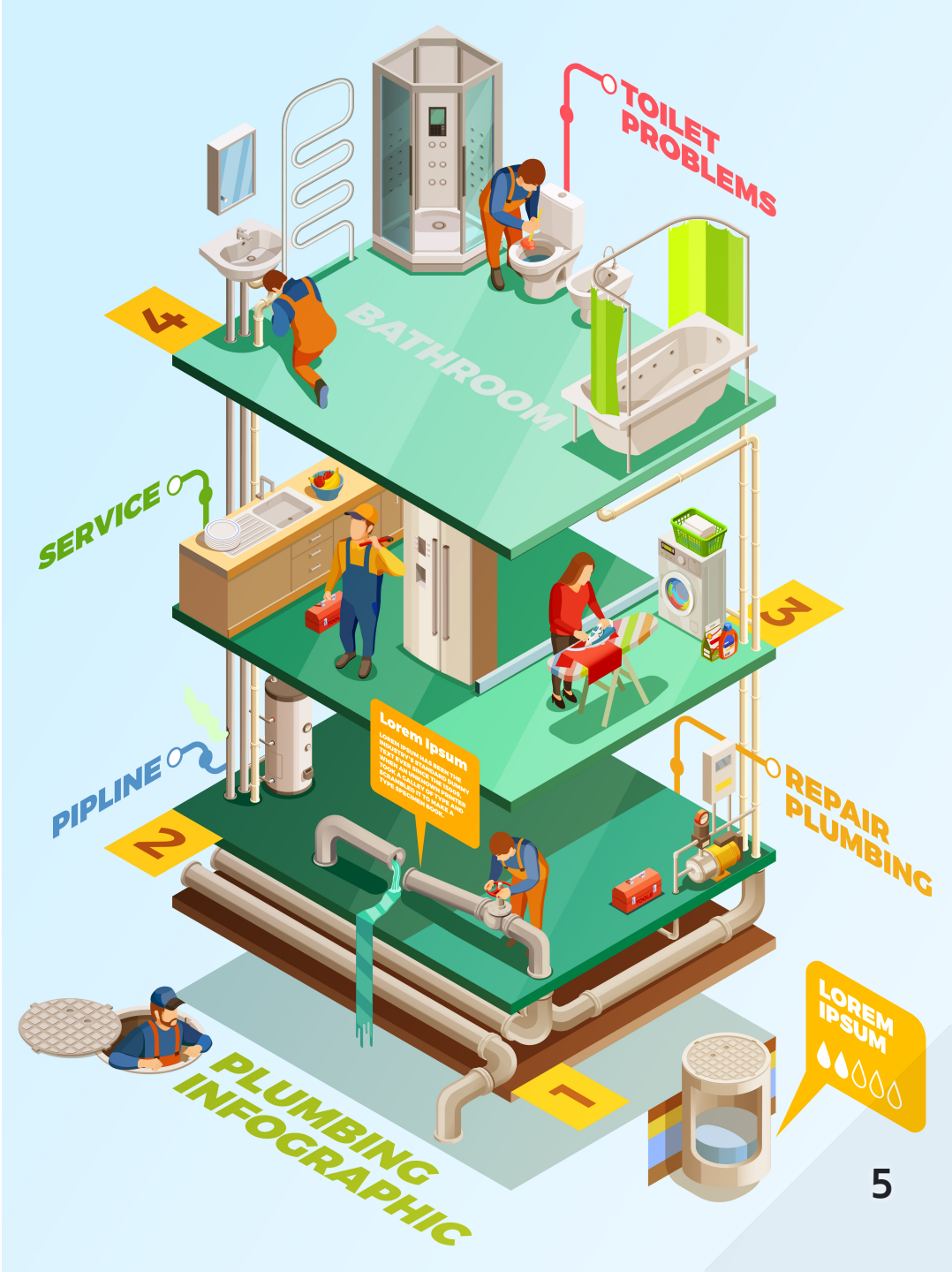


Energy - Power Grid

[Web Site]

- User Consumption
  - Always
- Grid Production
  - Our Problem
- Resiliency
  - Our Backups Have Backups (Hopefully)
- **Focus**
  - **Keeping the Lights On (Literally)**

Image by macrovector on Magnific
[Equivalent energy image not found]



Energy - Solar/Wind

[Mobile App]

- Energy Production
  - Local
- Grid Consumption
  - Backup
- Grid Production
  - Their Problem
- Resiliency
  - Battery, Power Grid
- **Focus**
  - **Smartly Using and Storing Energy**

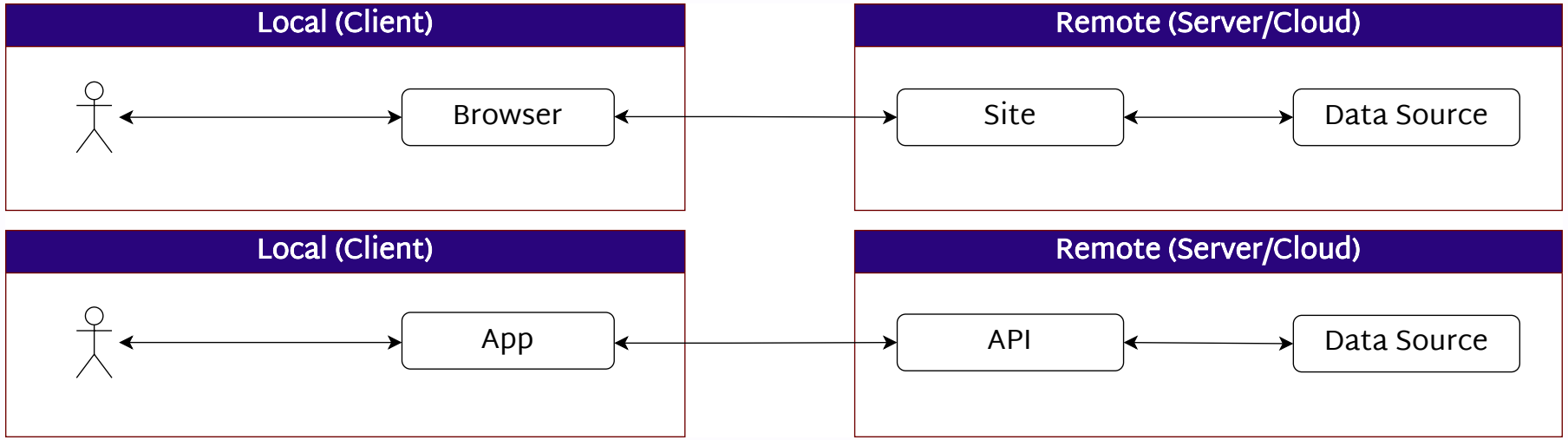
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Part 2

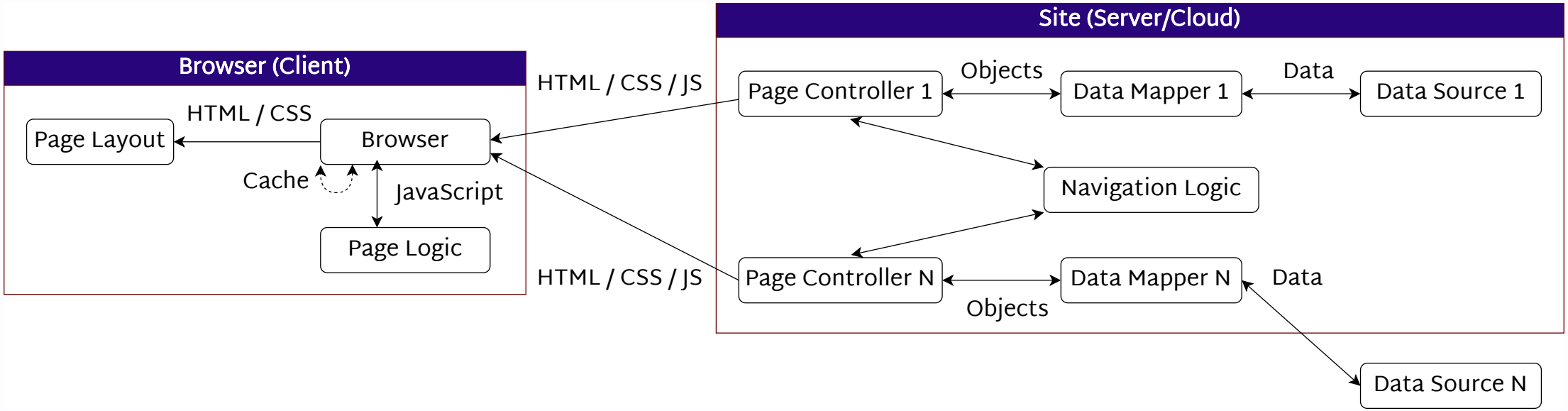
Mobile is just like Web but smaller, right!?

Web vs Mobile - High Level



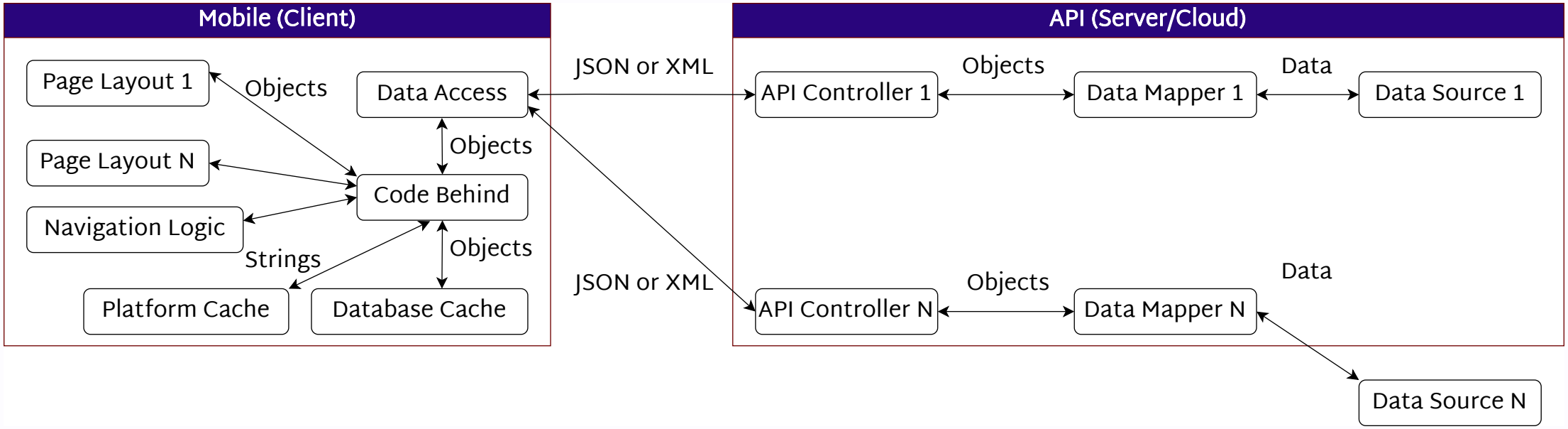
Same thing, right? Problem solved! Crisis averted!
Thank you and good night!

Web Architecture - Components



- Server/cloud stack is the focus here, turning data into pages and layouts
- Complex layouts are generally built on top of templating frameworks
- **Web devs expect fat data pipes in the server/cloud stack to get large data payloads (object graphs) to choose what to filter/display in page layouts**

Mobile Architecture - Components



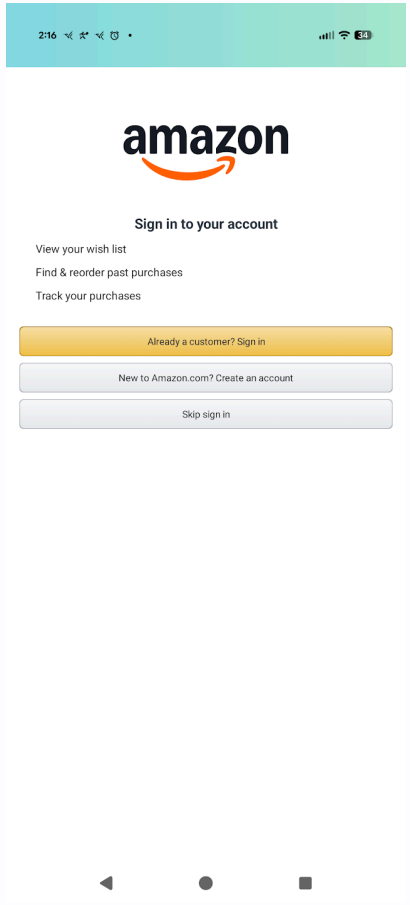
- App handles interactivity, page layouts, local caching, and API calls
- API stack delivers only data or status codes in response to app requests
- App pages tend to be focused on single tasks that call the API selectively
- **Mobile/API devs focus more on reliable, just-in-time data delivery**

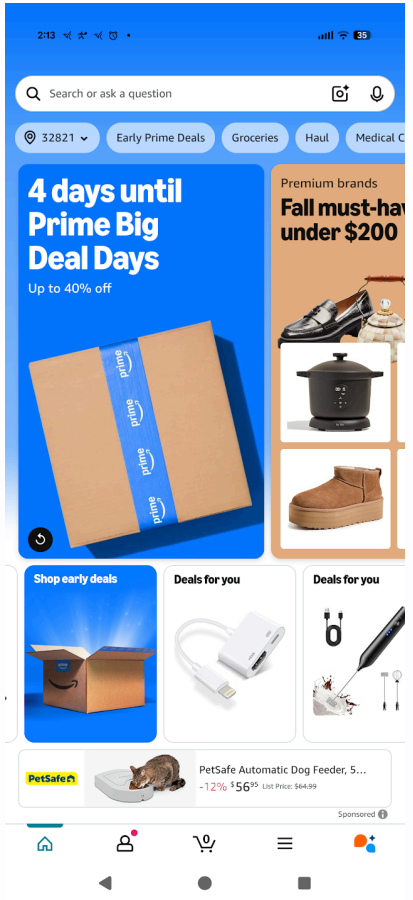
Part 3

Being of two minds about data

Mobile Apps - Platform Cache

- Tokens are often created on login, cached locally, and renewed in the background
- Apps often store the most recent token in fast secure platform-level key:value cache
- If a valid token exists on launch, the app can skip directly to a home page





Mobile Apps - Database Cache

- Akavache - github.com/reactiveui/Akavache
  An asynchronous typed SQLite cache for cross-platform applications

```
// Store an object
var user = new User { Name = "John", Email = "john@example.com" };
await CacheDatabase.UserAccount.InsertObject("current_user", user);

// Retrieve an object
var cachedUser = await CacheDatabase.UserAccount.GetObject<User>("current_user");

// Store with expiration
await CacheDatabase.LocalMachine.InsertObject("temp_data",
  someData, DateTimeOffset.Now.AddHours(1));

// Get or fetch pattern
var data =
  await CacheDatabase.LocalMachine.GetOrFetchObject("api_data",
    async () => await httpClient.GetFromJsonAsync<ApiResponse>("https://api.example.com/data"));
```

Part 4

You got Presentation Logic on my Business Logic

You got Business Logic in my Presentation Logic

MVVM Pattern - Theory

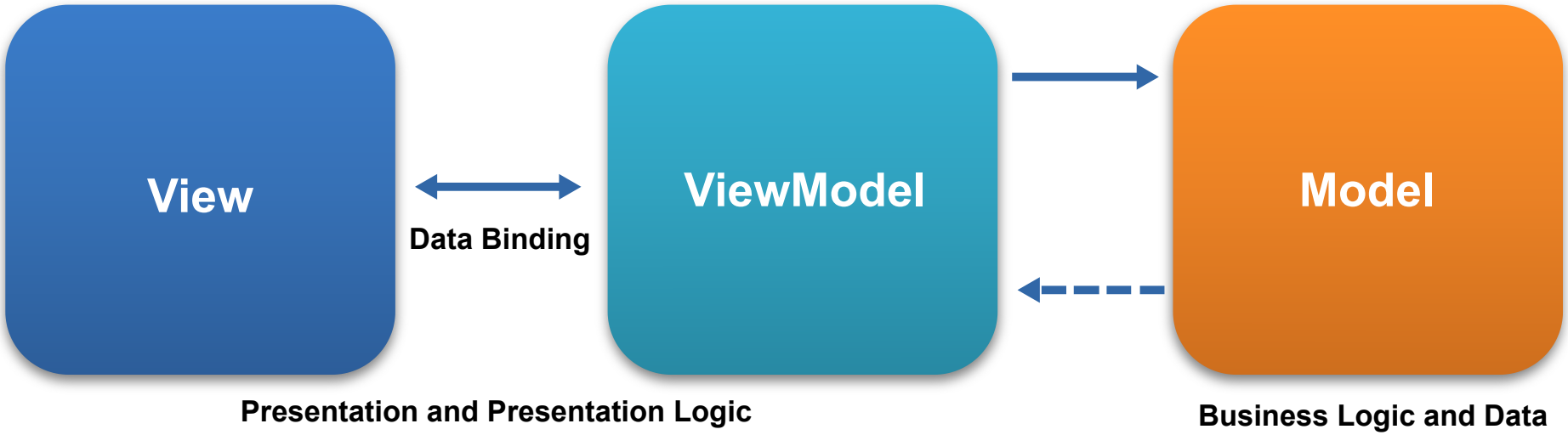
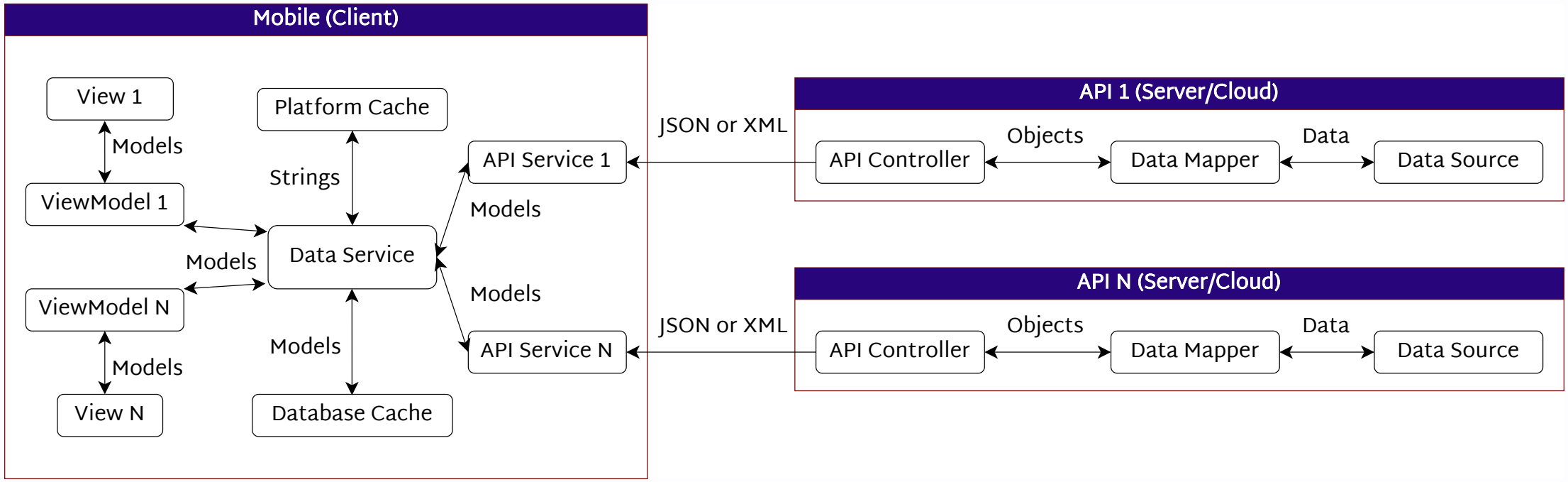


Image from Wikipedia

Mobile Architecture - Real-world

Component	Architecture	Purpose
Page Layouts	Views	Presentation
Navigation Logic	ViewModels	Navigation Router
Code Behind	ViewModels	Presentation Logic
Code Behind	ViewModels	Data Manipulation Logic
Local Caches	Data Service	Data Source Logic
Data Access	API Services	Data Translation Logic
Data Access	Models	Internal Data Definitions
Data Access	Data Transfer Objects	External Data Definitions

Mobile Architecture - Network

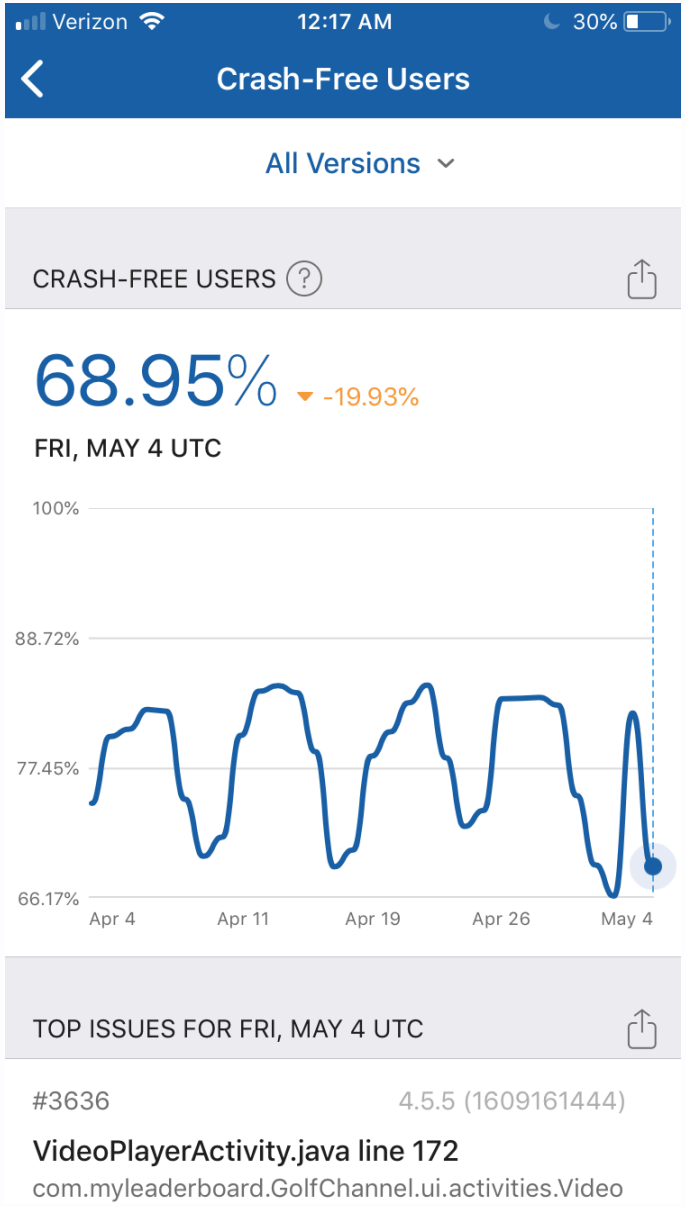


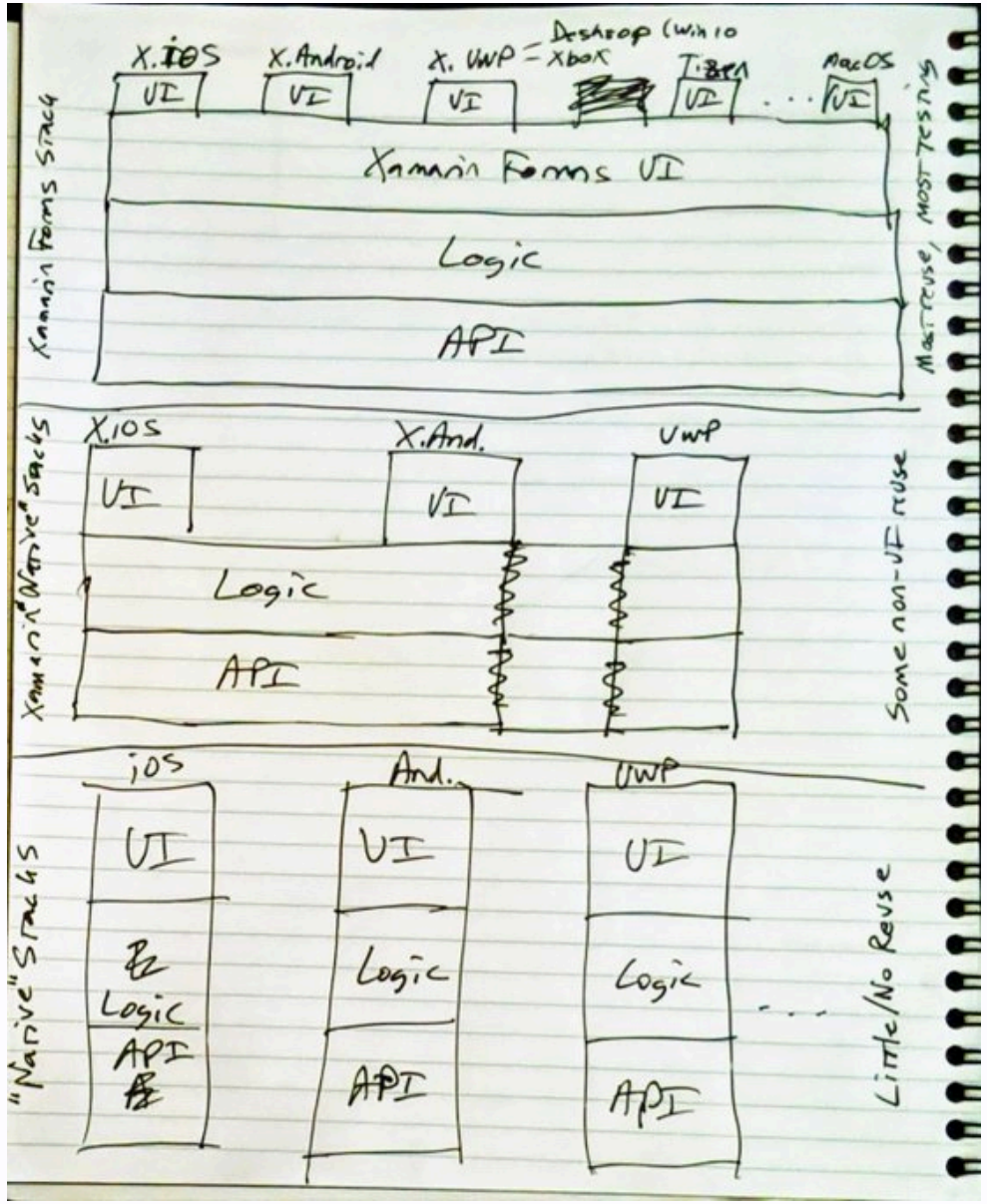
Part 5

Why?

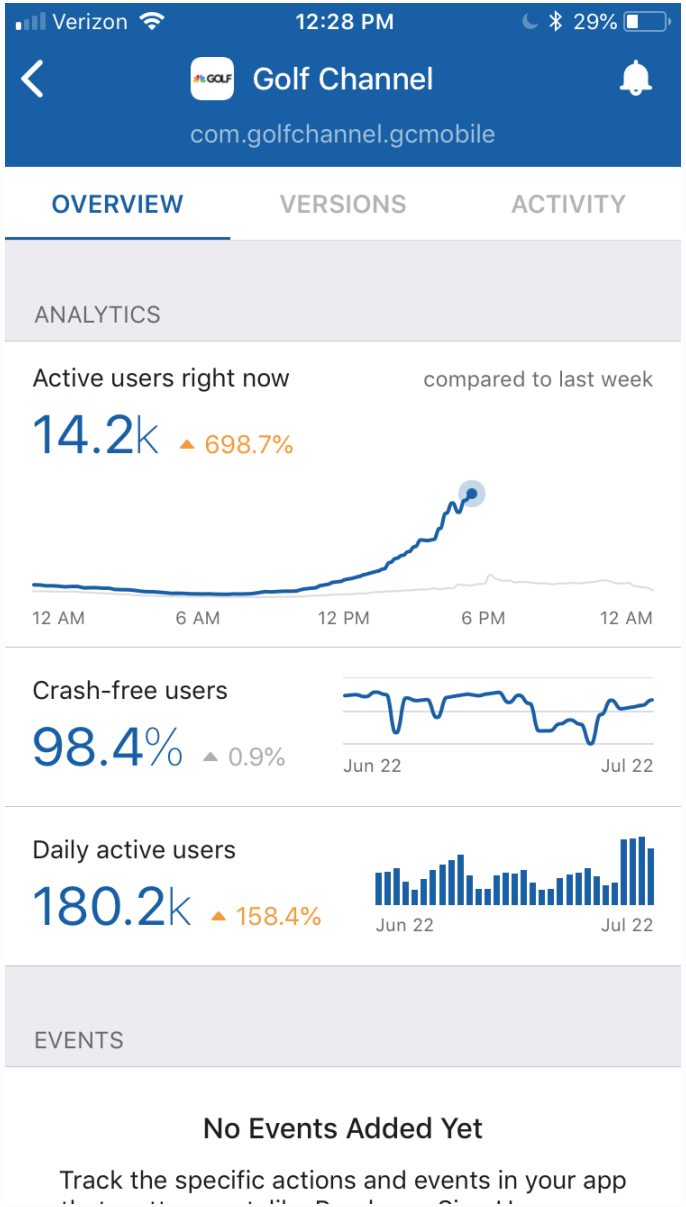

11:14 PM
GOLF
PRESENTED BY
TaylorMade
Tiger Woods


11:14 PM
GOLF
PRESENTED BY
TaylorMade
Tiger Woods


Verizon
12:17 AM
30%
Crash-Free Users
All Versions
CRASH-FREE USERS
68.95%
-19.93%
FRI, MAY 4 UTC
100%
88.72%
77.45%
66.17%
Apr 4
Apr 11
Apr 19
Apr 26
May 4
TOP ISSUES FOR FRI, MAY 4 UTC
#3636
4.5.5 (1609161444)
VideoPlayerActivity.java line 172
com.myleaderboard.GolfChannel.ui.activities.Video


X.iOS
X.Android
X. UWP = XBox
Desktop (Win10
MacOS
VI
Xamarin Forms UI
Logic
API
Xamarin Forms Stack
Most reuse
X.iOS
X.And.
UWP
Logic
API
Xamarin "Native" Stacks
Some non-UI reuse
iOS
And.
UWP
UI
Logic
API
"Native" Stacks
Little/No Reuse


Verizon LTE 6:14 PM 44%
GOLF
PRESENTED BY
TaylorMade
Tiger Woods
MASTERS CHAMPION
Team TaylorMade


Verizon 12:28 PM 29%
Golf Channel
com.golfchannel.gcmobile
OVERVIEW
VERSIONS
ACTIVITY
ANALYTICS
Active users right now
compared to last week
14.2k ▲ 698.7%
12 AM
6 AM
12 PM
6 PM
12 AM
Crash-free users
98.4% ▲ 0.9%
Jun 22
Jul 22
Daily active users
180.2k ▲ 158.4%
Jun 22
Jul 22
EVENTS
No Events Added Yet
Track the specific actions and events in your app

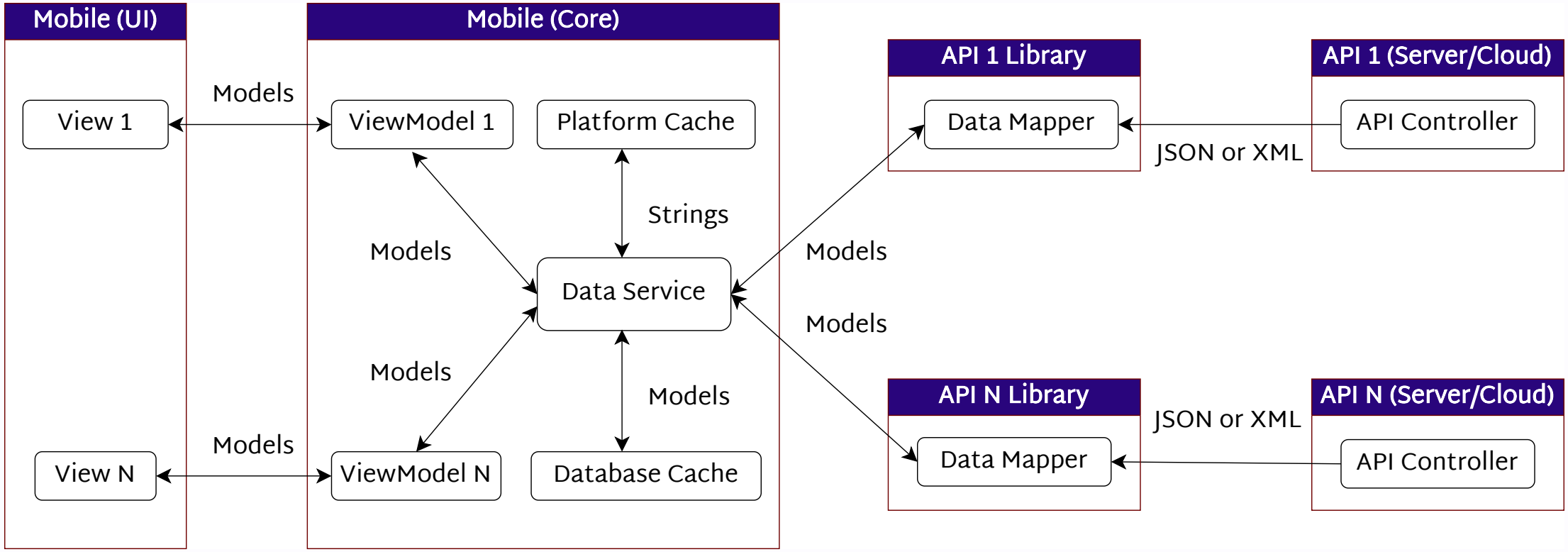
Part 6

How?

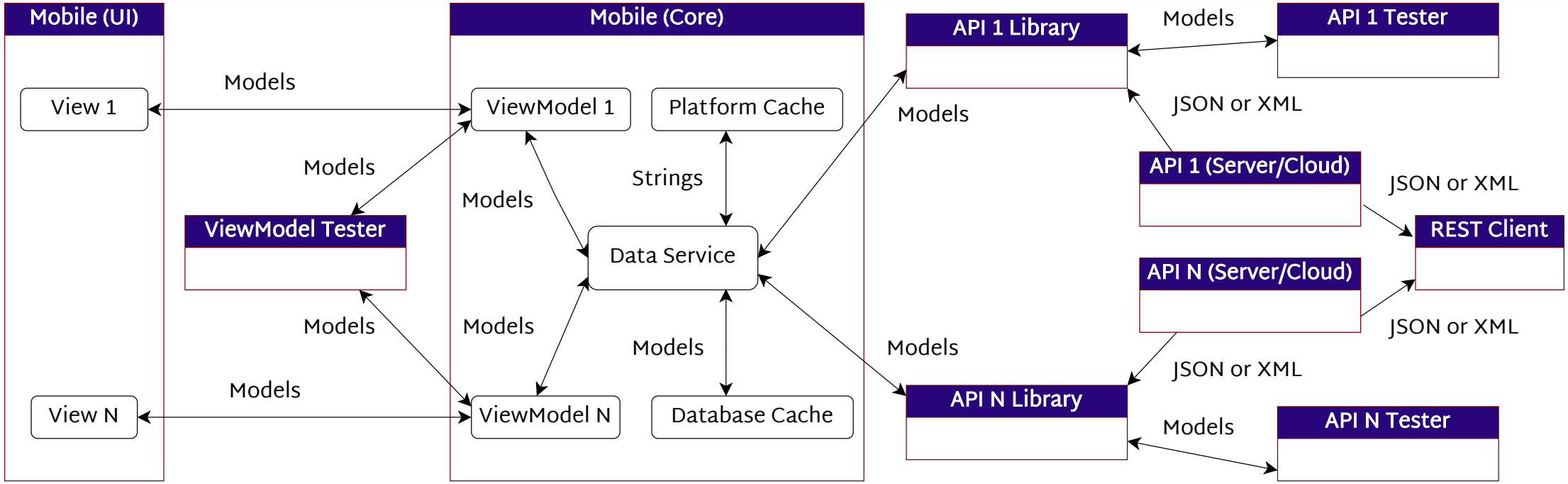
Mobile Architecture - Components

- ViewModels
  - Moving navigation out of the View allows ViewModels to be tested individually with frameworks like ReactiveUI (RxUI)
- Api Services
  - Individually testable to return app Models and not just Data Transfer Objects (DTOs)
- Data Service
  - Decides whether to call the local cache DB, the platform cache, an API endpoint, or a service based on the requested data and the app lifecycle

Mobile Architecture - Solution



Mobile Architecture - Testing



API Calls - Auto-generated from Interface

```
public interface IGitHubApi
{
    [Get("/users/{user}")]
    Task<User> GetUser(string user,
      [Authorize("Bearer")] string token);
}

var gitHubApi = RestService.For<IGitHubApi>("https://api.github.com");

// Will add the header "Authorization: Bearer OAUTH-TOKEN}" to the request
var octocat = await gitHubApi.GetUser("octocat", "OAUTH-TOKEN");
```

Part 7

There is another way ...

Problem Details

Pros

- Extensible JSON object
- Provides context to status code
- Can report errors or warnings
- Supported in .NET without ASP.NET
- Returned by Refit's ApiException

Cons

- Buggy return type in ASP.NET
- .NET implementations follow JSON

```
# NOTE: '\' line wrapping per RFC 8792
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "An RFC 7807 problem object",
  "type": "object",
  "properties": {
    "type": {
      "type": "string",
      "format": "uri-reference",
      "description": "A URI reference that identifies the problem type."
    },
    "title": {
      "type": "string",
      "description": "A short, human-readable summary of the problem type."
    },
    "status": {
      "type": "integer",
      "description": "The HTTP status code generated by the origin server \
for this occurrence of the problem.",
      "minimum": 100,
      "maximum": 599
    },
    "detail": {
      "type": "string",
      "description": "A human-readable explanation specific to this occurrence \
of the problem."
    },
    "instance": {
      "type": "string",
      "format": "uri-reference",
      "description": "A URI reference that identifies the specific occurrence \
of the problem. It may or may not yield further information if dereferenced."
    }
  }
}
```

Source: IETF RFC 9457 - Problem Details for HTTP APIs www.rfc-editor.org/rfc/rfc9457.html

Problem Details - Purchase Order Error

```
POST /purchase HTTP/1.1
Host: store.example.com
Content-Type: application/json
Accept: application/json, application/problem+json

{
  "item": 123456,
  "quantity": 2
}
```

```
HTTP/1.1 403 Forbidden
Content-Type: application/problem+json
Content-Language: en

{
 "type": "https://example.com/probs/out-of-credit",
 "title": "You do not have enough credit.",
 "detail": "Your current balance is 30, but that costs 50.",
 "instance": "/account/12345/msgs/abc",
 "balance": 30,
 "accounts": ["/account/12345",
              "/account/67890"]
}
```

Problem Details - Type Validation Error

```
POST /details HTTP/1.1
Host: account.example.com
Content-Type: application/json
Accept: application/json, application/problem+json

{
  "age": 42.3,
  "profile": {
    "color": "yellow"
  }
}
```

```
HTTP/1.1 422 Unprocessable Content
Content-Type: application/problem+json
Content-Language: en

{
 "type": "https://example.net/validation-error",
 "title": "Your request is not valid.",
 "errors": [
             {
               "detail": "must be a positive integer",
               "pointer": "#/age"
             },
             {
               "detail": "must be 'green', 'red' or 'blue'",
               "pointer": "#/profile/color"
             }
           ]
}
```

Part 8

References / Questions?

References

- Reactive Extensions (Rx or Rx.NET) - reactivex.io
  An API for asynchronous programming with observable streams
- ReactiveUI (RxUI) - www.reactiveui.net
  A composable Model-View-ViewModel framework for all .NET platforms!
- Refit - reactiveui.github.io/refit
  An automatic type-safe REST library for .NET Core, Xamarin and .NET
- Akavache - github.com/reactiveui/Akavache
  An asynchronous Key-Value store for cross-platform applications
- Splat - github.com/reactiveui/splat
  A cross-platform library for service location, unit testing, and logging